[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

LATHE, VERTICAL-BED, HORIZONTAL-SPINDLE, AND TURNING CENTER, VERTICAL-BED, HORIZONTAL-SPINDLE, COMPUTER NUMERIC CONTROL (CNC)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers computer numeric control (CNC), horizontal-spindle, vertical-bed lathes and turning centers intended for use in turning, boring, drilling, reaming, tapping, and facing workpieces of cast iron, steel, titanium, aluminum, graphite epoxy, and composite materials.

2. CLASSIFICATION. The vertical-bed lathes and turning centers shall be of the following types, swings, and sizes. The type, swing, and size to be furnished shall be as specified (see 7.2(b)).

Type I – Lathe, vertical-bed
Type II – Turning center, vertical-bed (C-axis indexable live-tooling spindle)

Swing – Swings over bed are shown in table I

Size – Sizes are shown in table I

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3416

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TABLE I. Swings and sizes of vertical-bed lathes and turning centers.

Swing diameter over bed, inches	Swing diameter over cross-slide, inches	Spindle throughfeed stock diameter, inches	Spindle drive motor, continuous horsepower	Size, center-to-center distance, inches										
				4	12	14	20	22	24	30	36	40	50	80
6	1.7	1.50	10	X										
8	3	1.75	10	X										
12	8	1.75	10		X									
14	10	1.75	10			X		X						
16	16	1.75	10			X								
20	10	1.75	25				X				X	X		
22	21	3.00	25						X	X		X		
24	12	3.50	35								X		X	X
28	17	4.00	60						X		X	X		X
32	25	7.50	75				X							

Note 1: An X denotes available swing and size combinations.

Note 2: The lathe or turning center size and capacity shall not be less than the stated requirements selected from table I for the swing and size specified. The ranges of size and capacity shown in table I imply that the lathe or turning center offered shall be equal to or larger than the size and capacity stated but smaller than the next available size and capacity. When the largest lathe or turning center shown in table I is selected, the size and capacity of the lathe or turning center offered shall not exceed the stated requirements by more than 10 percent.

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3. SALIENT CHARACTERISTICS

3.1 General requirements. The vertical-bed lathe or turning center shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. The lathe or turning center shall be constructed from materials with sufficient mass and strength and configured in shape and size that shall support the workpiece and machine components while minimizing element deflections and initiated and/or transmitted vibration during operation. The rotational axis of the workpiece shall be horizontal. The longitudinal axes of the machineways shall be parallel to the rotational axis of the workpiece and shall support the carriage(s), tailstock, steady rests, or tool rests in a plane that is perpendicular to the horizontal plane. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Access doors or removable panels shall be provided for servicing all components located within the machine.

3.1.1 Motion control axes. Motion control axes of the vertical-bed lathe or turning center are shown in figure 1 and shall be provided as specified (see 7.2(c)). Axis feed rate ranges and rapid traverse rates shall be as specified (see 7.2(d)).

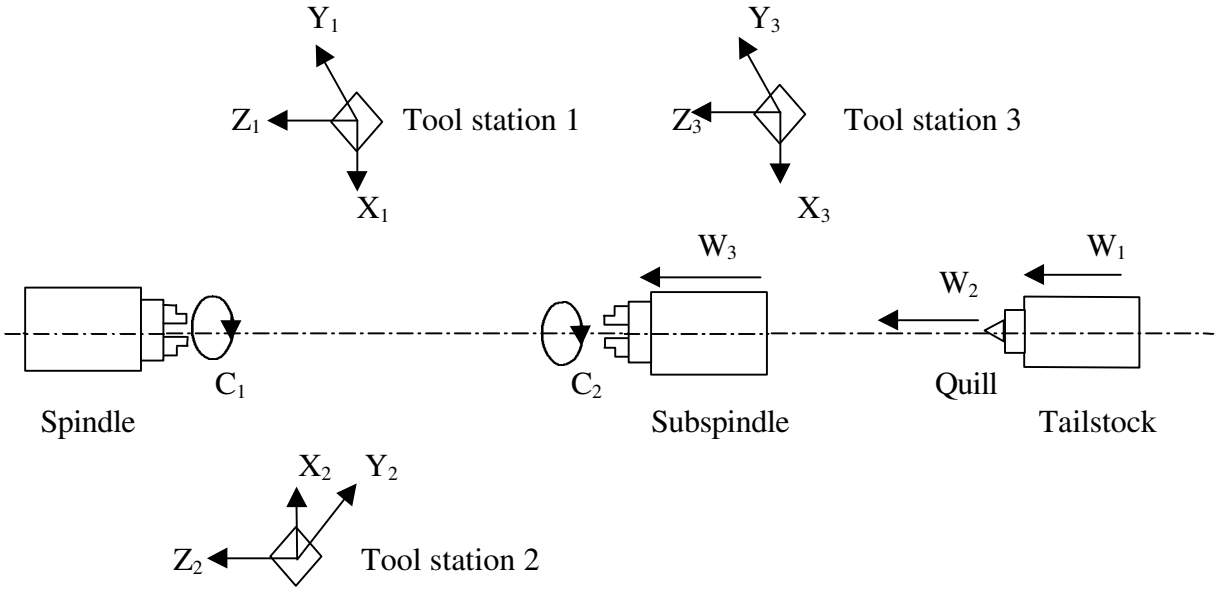


FIGURE 1. Identification of vertical-bed lathe and turning center motion control axes.

3.2 Components. The lathe or turning center shall include, as a minimum, the following components:

3.2.1 Base. The base shall support the lathe or turning center bed and all components mounted thereon. The base shall provide a means for leveling and anchoring the machine. A swarf collection area shall be provided either adjacent to or within the base to collect and retain turning chips and coolant fluid.

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3.2.2 Bed. The vertical bed shall support the machineways upon which the carriage(s), subspindle, tailstock, and tool or workpiece rests travel. The bed shall be configured to allow the chip swarf to fall directly into the collection area.

3.2.3 Headstock. The headstock shall enclose and support the spindle and spindle drive.

3.2.3.1 Spindle. The spindle shall be fabricated from alloy steel, then hardened and ground. The spindle shall be supported on preloaded, antifriction roller or ball bearings within the headstock. All spindle bearings shall be sealed and lubricated for life. The spindle shall be dynamically balanced both before and after assembly. The spindle shall be thermally isolated from the headstock and spindle drive. Unless otherwise specified (see 7.2(e)), the spindle nose shall conform to American National Standards Institute (ANSI) B5.9. Unless otherwise specified (see 7.2(f)), the manufacturer's standard specifications shall apply to the following:

a. Spindle speed range.
b. Spindle nose type and size.
c. Spindle taper type and size.
d. Spindle chuck type and size.
e. Spindle throughfeed maximum stock diameter.

3.2.3.2 Spindle drive system. The spindle drive system shall supply the necessary power to rotate the workpiece at the selected speed. Unless otherwise specified (see 7.2(g)), the spindle drive motor horsepower (continuous rating) and speed and drive system speed step changes shall be that considered by the manufacturer as standard.

3.2.3.3 Live-tooling spindle. The type II turning center shall be provided with a C-axis live-tooling spindle having incremental angular indexing over a range of 360 degrees as specified (see 7.2(h)).

3.2.4 Machineways. Unless otherwise specified (see 7.2(i)), machineways shall be of the design considered by the manufacturer as standard. A means shall be provided to prevent coolant and chip swarf from accumulating on machineways and becoming lodged between sliding parts.

3.2.5 Carriage. The lathe or turning center shall have one carriage for each Z-axis specified. The carriage(s) shall be mounted on the bed machineways, shall move parallel to the rotational axis of the workpiece, and shall support the cross-slide(s).

3.2.6 Cross-slide. The lathe or turning center shall have a cross-slide mounted on each carriage. The cross-slide shall travel perpendicular to the rotating axis of the workpiece in a vertical plane.

3.2.7 Tool station(s). The tool station(s) shall be mounted on the cross-slide(s) of the lathe or turning center. The tool station(s) shall position the selected cutting tool at the appropriate place, angle, and attitude for the desired process. The lathe or turning center shall be equipped with the number and type of tool stations (e.g., tool post, multitool turret, or gang tooling) as specified (see 7.2(j)).

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3.2.7.1 Tool turret(s). When a tool turret(s) is provided, the following items for each turret shall be as specified (see 7.2(k)):

a. Style (e.g., drum, disc, or carousel).
b. Location.
c. Number of fixed tool sites.
d. Number and type of power (live) tool sites.
e. Size of the tool site.

3.2.7.2 Gang tooling. When gang tooling is provided, the following items shall be as specified (see 7.2(l)):

a. Location.
b. Number of fixed tool sites.
c. Number and type of power (live) tool sites.
d. Size of the tool site.

3.2.8 Axis drive mechanisms. Each motion control axis of the lathe or turning center shall be actuated by an independent drive that shall have a drive motor, an actuation mechanism, and a closed-loop feedback control system to position the driven component. Unless otherwise specified (see 7.2(m)), all axis drive motors shall be infinitely variable-speed, AC servomotors.

3.2.9 Subspindle assembly. Unless otherwise specified (see 7.2(n)), the type II turning center shall have a subspindle assembly located at the end of the bed opposite of the headstock. The subspindle assembly shall enclose and support the subspindle and the subspindle drive.

3.2.9.1 Subspindle. The centerline of the subspindle shall be coaxial with the centerline of the headstock spindle and shall be capable of synchronous rotation with the headstock spindle. The subspindle shall be fabricated from alloy steel, then hardened and ground. The subspindle shall be supported on preloaded, antifriction roller or ball bearings that are sealed and lubricated for life. The subspindle shall be dynamically balanced before and after assembly. Unless otherwise specified (see 7.2(o)), the subspindle nose shall conform to ANSI B5.9. Unless otherwise specified (see 7.2(p)), the manufacturer's standard specifications shall apply to the following:

a. Subspindle speed range.
b. Subspindle nose type and size.
c. Subspindle taper type and size.
d. Subspindle chuck type and size.
e. Subspindle throughfeed maximum stock diameter.

3.2.9.2 Subspindle drive. Unless otherwise specified (see 7.2(q)), the subspindle drive motor horsepower (continuous rating) and speed and drive system speed step changes shall be that considered by the manufacturer as standard.

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3.2.10 Tailstock. When provided, the lathe or turning center shall have a tailstock located on the bed opposite of the headstock. The centerline of the tailstock shall be coaxial with the centerline of the headstock spindle. The tailstock shall be equipped with a moveable quill and be capable of accommodating various material-holding centers and cutting tools. The tailstock shall have a locking device to hold it in position after adjustment.

3.2.11 Chip conveyor system. If required, the lathe or turning center shall be supplied with a chip conveyor system as specified (see 7.2(r)). If a chip conveyor is provided, the discharge chute height shall be as specified (see 7.2(s)). The conveyor system shall remove and convey the turning chips away from the swarf collection area to a chip collection bin located adjacent to the lathe or turning center. The conveyor system shall function as a complete system and shall interface with the lathe or turning center and the CNC system.

3.2.12 Safety devices. An automatic stop circuit shall be provided that will stop the operation of the spindle, the subspindle (when provided) and all axis motion drives if any motor becomes overloaded, overheated, or exceeds any operating parameter. The spindle, the subspindle (when provided), and all axis drives shall automatically switch to manual servocontrol during all safety interruptions. Limit switches shall be provided at both ends of each axis of travel to prevent overtravel. Automatic machine-actuated safety warnings, trips, and/or stoppages will activate an indicator light(s) on the operator control panel and produce an audible alarm. When specified (see 7.2(t)), torque-limiting devices shall be included on all axis drives. The lathe or turning center shall have a protective enclosure and splash panels considered by the manufacturer as standard for the machine. The enclosure shall be configured to afford easy operator access to and observation of the workpiece.

3.2.13 Control system. The lathe or turning center shall be capable of manual control from a local operator station, semiautomatic control from manual data input (MDI) devices, and automatic control from instructions stored in the memory of a CNC system based on either a personal computer (PC) open-architecture or other software-programmable microprocessor design. Unless otherwise specified (see 7.2(u)), the control system components shall be industrial grade, and enclosures shall be type 12 in accordance with the National Electrical Manufacturers Association (NEMA) 250. Unless otherwise specified (see 7.2(v)), the lathe or turning center shall be capable of operating in the following modes:

a. Manual operation mode.
b. Manual data input (MDI) mode.
c. Programming mode.
d. Automatic mode
e. Background editing mode.
f. Machine diagnostic/maintenance/monitoring modes.

3.2.13.1 Machine status indicators. Unless otherwise specified (see 7.2(w)), the lathe or turning center shall include, as a minimum, the following machine status indicators:

a. Power on indicator.

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b. Operational mode indicator.
c. Spindle power indicator.
d. Spindle speed indicator.
e. Axis movement direction indicators.
f. Axis movement speed indicators.
g. Subspindle power indicator, when subspindle is provided.
h. Subpindle speed indicator, when subspindle is provided.
i. Accessory system (e.g., air, hydraulic, coolant, lubrication) status indicator(s).
j. Overload and/or trouble indicator.
k. Overload and/or trouble audible alarm.

3.2.13.2 Manual controls. Unless otherwise specified (see 7.2(x)), the lathe or turning center shall include, as a minimum, the following manual switches and controls:

a. Emergency stop.
b. Key-actuated system lock.
c. Power on/off.
d. Spindle start/stop.
e. Spindle speed override.
f. Spindle incremental angular indexing, when a live-tooling spindle is provided.
g. Spindle drive gear selection, when a multispeed gear-drive transmission is provided.
h. Carriage motion.
i. Cross-slide motion.
j. Tool station indexing when a multitool station(s) is provided.
k. Subspindle start/stop, when a subspindle is provided.
l. Subspindle speed override, when a subspindle is provided.
m. Subspindle incremental angular indexing, when a live-tooling subspindle is provided.
n. Subspindle drive gear selection, when a multispeed gear-drive transmission is provided.
o. Tailstock motion, when a tailstock is provided.
p. Quill motion, when a quill is provided.
q. Feed hold.
r. Feed rate override.
s. Jog motion and jog direction.
t. Chip conveyor system, when provided.
u. CNC system start/stop (cycle start/stop).

3.2.13.3 Foot, hand-held, and pendant controls. If required, in addition to the standard control panel, the lathe or turning center shall have separate foot, hand-held, or pendant controls as specified (see 7.2(y)). The foot, hand-held, or pendant controls shall be easily accessible within the operator work zone. These controls shall allow the operator to override the CNC system and manually control the machine functions.

3.2.13.4 Computer numeric control. The lathe or turning center shall have a CNC system that is a completely integrated, functional unit. The CNC system shall operate as a continuous path system providing full contouring capability. The term “full contouring”, as used herein, shall be defined

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as a closed-loop control system having simultaneous control of position, direction, and rate of movement of all controlled axes. The CNC system shall always be active when electric power is supplied to the lathe or turning center. Unless otherwise specified (see 7.2(z)), the CNC system shall, as a minimum, automatically control the following functions:

a. Spindle speed.
b. Spindle drive gear selection, when a multispeed gear-driven spindle is provided.
c. Spindle incremental angular indexing, when a live-tooling spindle is provided.
d. Carriage motion.
e. Cross-slide motion.
f. Tool station indexing, when a multitool station(s) is provided.
g. Subspindle speed, when a subspindle is provided.
h. Subspindle drive gear selection, when a multispeed gear-driven spindle is provided.
i. Subspindle incremental angular indexing, when a live-tooling subspindle is provided.
j. Tailstock motion, when a tailstock is provided.
k. Quill motion, when a quill is provided.
l. Chip conveyor system, when provided.
m. All accessory systems (e.g., air, hydraulic, coolant, lubrication), when provided.
n. Other machine operating functions considered by the manufacturer as standard.

3.2.13.5 Hardware features. Unless otherwise specified (see 7.2(aa)), the CNC system shall include, as a minimum, the following hardware features:

a. A 32-bit microprocessor having a minimum operating speed of 16 megahertz and an internal protocol via a 32-bit local bus.
b. Buffer storage capacity sufficient to allow continuous program execution without non-programmed stops, delays, or pauses.
c. A 3.5-inch, 1.4-megabyte hard shell floppy disk drive.
d. An Institute of Electrical and Electronics Engineers (IEEE) 1284 parallel port.
e. An Electronic Industries Association (EIA) RS-232 serial port.
f. A 9-inch monochrome monitor.
g. Manual data input (MDI) device consisting of either a personal computer (PC) keyboard or an alphanumeric keypad with dedicated keys for specific machine functions.

3.2.13.6 Software features. Unless otherwise specified (see 7.2(bb)), the CNC system shall include, as a minimum, the following programming software and control features:

a. EIA RS-274 variable data block programming format (standard G-, M-, and T-codes).
b. EIA RS-494 programming format (32-bit binary and 7-bit ASCII).
c. Decimal point programming.
d. Conversational mode programming.
e. Inch/metric programming.
f. Programmable zero-point reference.
g. Absolute and incremental motion control programming.
h. Simultaneous control of all motion axes.

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i. Linear, circular, and helical motion interpolation.
j. Constant surface speed capability.
k. Manufacturer's standard suite of preprogrammed machine operational cycles.
l. Workpiece graphic display.
m. Tool path graphic display.
n. Tool length compensation.
o. Tool offsets.
p. Scale feedback compensation.
q. Drive backlash compensation.
r. Maintenance monitoring package.
s. Diagnostics assistance.

3.2.14 Lubrication system. Unless otherwise specified (see 7.2(cc)), an automatic lubrication system shall be provided to direct lubricant from a central reservoir to all machine components requiring continuous lubrication. The automatic lubrication system shall include a reservoir, piping, valves, pump, motor, strainer, heat exchanger, and a cleanable or replaceable filter. A means shall be provided for determining the oil level in the reservoir. Oil high-temperature and low-level alarms shall be provided. Components requiring periodic manual lubrication shall be readily accessible through doors or removable panels. When a spindle or subspindle gear-drive transmission is provided, a separate lubrication system for the gears shall be furnished.

3.2.15 Coolant system. Unless otherwise specified (see 7.2(dd)), an automatic cutting tool flood coolant system shall be provided that shall direct a selectable controlled flow rate of coolant to the cutting tool. The cooling system shall include a reservoir, piping, valves, pump, motor, heat exchanger, and a cleanable or replaceable filter. A means shall be provided for determining the coolant level in the reservoir. Coolant high-temperature and low-level alarms shall be provided.

3.2.16 Hydraulic system. If required for component operation, a hydraulic system shall be provided as specified (see 7.2(ee)). When provided, the hydraulic system shall conform to the National Fluid Power Association (NFPA) T2.24.1. The hydraulic system shall include a reservoir, piping, valves, pump, motor, strainer, heat exchanger, and a cleanable or replaceable filter. A means shall be provided for determining the fluid level in the reservoir. Hydraulic oil high-temperature, high-pressure, low-pressure, and low-level alarms shall be provided. An identification plate or chart shall be attached to the lathe or turning center stating the type and Society of Automotive Engineers (SAE) identification number of the hydraulic fluid used.

3.2.17 Compressed air system. If required for operation of pneumatically actuated components, a compressed air system connected either to shop air or to a dedicated air compressor and tank shall be provided as specified (see 7.2(ff)). When provided, the system shall supply compressed air at the pressure and flow rate as specified (see 7.2(gg)). The compressed air system shall include piping, valves, fittings, dryer, a cleanable or replaceable filter, and other appurtenances necessary for the operation of pneumatic components. The compressed air system shall conform to the requirements of NFPA T2.25.1M.

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3.2.18 Electrical system. Unless otherwise specified (see 7.2(hh)), the electrical system shall conform to the National Fire Protection Association (NFPA) 79. The machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(ii)). Unless otherwise specified (see 7.2(jj)), the CNC system power shall be provided by a separate 120-volt, single-phase, 60-hertz circuit from the machine main power supply.

3.2.19 Electric motors. Spindle and subspindle drive motors and motion control motors shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to NEMA MG 1 and MG 7, respectively. Unless otherwise specified (see 7.2(kk)), spindle and subspindle drive motors shall have open dripproof enclosures. Motion and position control motors and controls shall conform to NEMA ICS 1 and NEMA ICS 3.

3.2.20 Optional accessories. Optional accessories shall be provided as specified (see 7.2(ll)).

3.3 Performance. When tested in accordance with sections 5.3 through 5.3.3, the lathe or turning center shall meet either the precision or standard performance category requirements shown in table II as specified (see 7.2(mm)).

TABLE II. Performance requirements.

Test	Parameter	Performance category	
		Precision	Standard
Linear displacement accuracy	All X, Y, and Z directions	0.0003 inch/10 inches	0.001 inch/10 inches
Bidirectional repeatability	All X, Y, and Z directions	±0.0002 inch	±0.0005 inch
Maximum parallelism error between spindle and subspindle C-axes and each Z-axis		0.0003 inch	0.0005 inch
Spindle dynamic error	Radial error-average	0.0003 inch	0.0005 inch
Spindle thermal growth		0.0015 inch	0.0025 inch
Spindle C-axis indexing accuracy		±1.8 arc seconds	±3.6 arc seconds
Spindle C-axis indexing repeatability		±1.0 arc second	±1.8 arc seconds
Turret indexing accuracy		±3 arc seconds	±5 arc seconds
Turret indexing repeatability		±1.5 arc seconds	±2.5 arc seconds
Contouring performance-10-inch ball bar test	100°, 190° and 360°	0.0005 inch	0.0015 inch

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(nn)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.6. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI B11.6 and OSHA standards, the OSHA standards shall apply.

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3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(oo)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(pp)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(qq)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/overcurrent rating.
g. Contract number or order number.
h. National stock number.
i. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.2(rr)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and

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quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The lathe or turning center shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(ss)), the lathe or turning center shall be tested in accordance with sections 5.3.1 through 5.3.3. Should a malfunction or a failure occur during any test, the problem shall be corrected, and the test shall be rerun from the beginning.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(tt)), the manufacturer shall be responsible for supplying all tooling and materials and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(uu)). Unless otherwise specified (see 7.2(vv)), the machine shall be tested in the environmental conditions considered by the manufacturer as standard for machine performance verification.

5.3.2 Operational test. The lathe or turning center shall be operated in accordance with the manufacturer's standard operating procedures. The lathe or turning center shall be operated for not less than 8 hours in which time proper operation of all controls and features shall be verified.

5.3.3 Performance tests. Unless otherwise specified (see 7.2(ww)), the performance tests shall be conducted in accordance with procedures considered by the manufacturer as standard for machine performance verification.

5.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.2(xx)).

5.4 Acceptance. Unless otherwise specified (see 7.2(yy)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the performance tests shall be compared with the performance requirements shown in table II. Failure of the machine to meet the performance requirements shown in table II shall be cause for rejection.

6. PACKAGING. Unless otherwise specified (see 7.2(zz)), preservation, packing, and marking shall be as specified in the contract or purchase order.

7. NOTES

7.1 Sources of referenced documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

American National Standards Institute (ANSI)

ANSI B5.9	Spindle Noses for Tool Room Lathes, Engine Lathes, Turret Lathes, and Automatic Lathes
ANSI B11.6	Safety Requirements for the Construction, Care, and Use of Lathes

Applications for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

Electronic Industries Association (EIA)

RS-232	Interface Between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Exchange
RS-274	Interchangeable Variable Block Data Format for Positioning and Contouring/Positioning Numerically Controlled Machines
RS-494	32 bit Binary CL (BCL) and 7 Bit ASCII CL (ACL) Exchange Input Format for Numerically Controlled Machines

Applications for copies should be sent to the Electronic Industries Association, 2500 Wilson Boulevard, Arlington, VA 22201-3834.

The Institute of Electrical and Electronics Engineers, Inc. (IEEE)

IEEE 1284	Standard Signaling Method for a Bidirectional Parallel Peripheral Interface for Personal Computers

Applications for copies should be sent to the Institute of Electrical and Electronics Engineers, Inc., 345 East 47th Street, New York, NY 10017.

National Electrical Manufacturers Association (NEMA)

NEMA ICS 1	Industrial Control and Systems General Requirements
NEMA ICS 3	Industrial Systems
NEMA MG 1	Motors and Generators
NEMA MG 7	Motion/Position Control Motors and Controls

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NEMA 250 Enclosures for Electrical Equipment (1000 Volts Maximum)

Applications for copies should be sent to the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

National Fire Protection Association (NFPA)

NFPA 79Electrical Standard for Industrial Machinery

Applications for copies should be sent to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.

National Fluid Power Association (NFPA)

NFPA T2.24.1	Hydraulic Fluid Power - Systems Standards for Stationary Industrial Machinery
NFPA T2.25.1M	Pneumatic Fluid Power - System Standard for Industrial Machinery

Applications for copies should be sent to the National Fluid Power Association, 3333 N. Mayfair Road, Milwaukee, WI 53222-3219.

Occupational Safety and Health Administration (OSHA)

OSHA 1910	Occupational Safety and Health Standards for General Industry, (29 CFR 1910)

Applications for copies should be sent to the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type, swing, and size of lathe or turning center required (see 2).
c. Motion control axes, as specified (see 3.1.1).
d. Motion axis feed rate ranges and rapid traverse rates, as specified (see 3.1.1).
e. Spindle nose, if different (see 3.2.3.1).
f. Spindle specifications, if different (see 3.2.3.1).
g. Spindle drive motor horsepower and speed, if different (see 3.2.3.2).
h. Live-tooling spindle incremental angular indexing, as specified (3.2.3.3).
i. Machineways, if different (see 3.2.4).
j. Number and type of tool stations, as specified (see 3.2.7).
k. Tool turret(s), as specified (see 3.2.7.1).
l. Gang tooling, as specified (see 3.2.7.2).

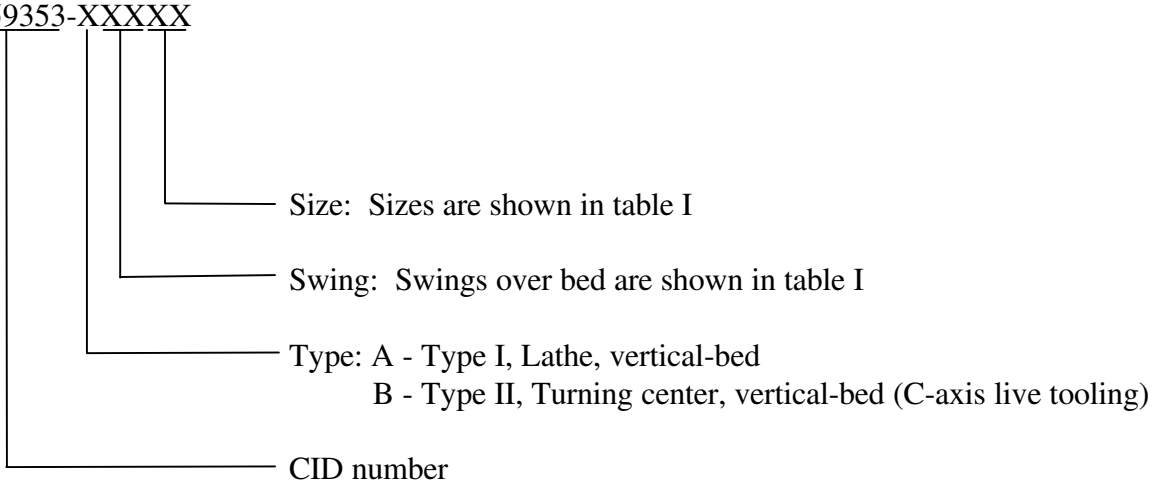
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m. Axis drive motors, if different (see 3.2.8).
n. Subspindle assembly, if different (see 3.2.9).
o. Subspindle nose, if different (see 3.2.9.1).
p. Subspindle specifications, if different (see 3.2.9.1).
q. Subspindle drive motor horsepower and speed, if different (see 3.2.9.2).
r. Chip conveyor system, as specified (see 3.2.11).
s. Discharge chute height, as specified (see 3.2.11).
t. Drive axis torque limiting devices, if required (see 3.2.12).
u. Control system components and enclosures, if different (see 3.2.13).
v. Operational modes, if different (see 3.2.13).
w. Machine status indicators, if different (see 3.2.13.1).
x. Manual controls, if different (see 3.2.13.2).
y. Separate foot, hand-held, and pendent controls, as specified (see 3.2.13.3).
z. CNC system automatic controls, if different (see 3.2.13.4).
aa. CNC system hardware features, if different (see 3.2.13.5).
bb. CNC system software features, if different (see 3.2.13.6).
cc. Automatic lubrication system, if different (see 3.2.14).
dd. Coolant system, if different (see 3.2.15).
ee. Hydraulic system, as specified (see 3.2.16).
ff. Compressed air system, as specified (see 3.2.17).
gg. Compressed air pressure and flow rate, as specified (see 3.2.17).
hh. Electrical system industrial standard, if different (see 3.2.18).
ii. Electrical system voltage, as specified (see 3.2.18).
jj. CNC system voltage, if different (see 3.2.18).
kk. Motor enclosures, if different (see 3.2.19).
ll. Optional accessories, as specified (see 3.2.20).
mm. Performance category, as specified (see 3.3).
nn. Machine dimensions and weight, as specified (see 3.4).
oo. Systems of units required (see 3.6).
pp. Dual system of units, if required (see 3.6).
qq. Nameplate, if different (see 3.7).
rr. Lubrication plate or chart, if required (see 3.8).
ss. Testing, if different (see 5.3).
tt. Test responsibility, if different (see 5.3.1).
uu. Certification of calibration, as specified (see 5.3.1).
vv. Test environmental conditions, if different (see 5.3.1).
ww. Performance tests, if different (see 5.3.3).
xx. Optional tests, as specified (see 5.3.4).
yy. Acceptance, if different (see 5.4).
zz. Packaging requirements, if different (see 6).

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7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

AA59353-XXXXX



MILITARY INTERESTS:

Custodians
Army - AL
Navy - SH
Air Force - 99

Reviewers
Army - AR
Navy - MC, YD1
Air Force - 84

PREPARING ACTIVITY:
DLA - GS

(Project No. 3416-0277)

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